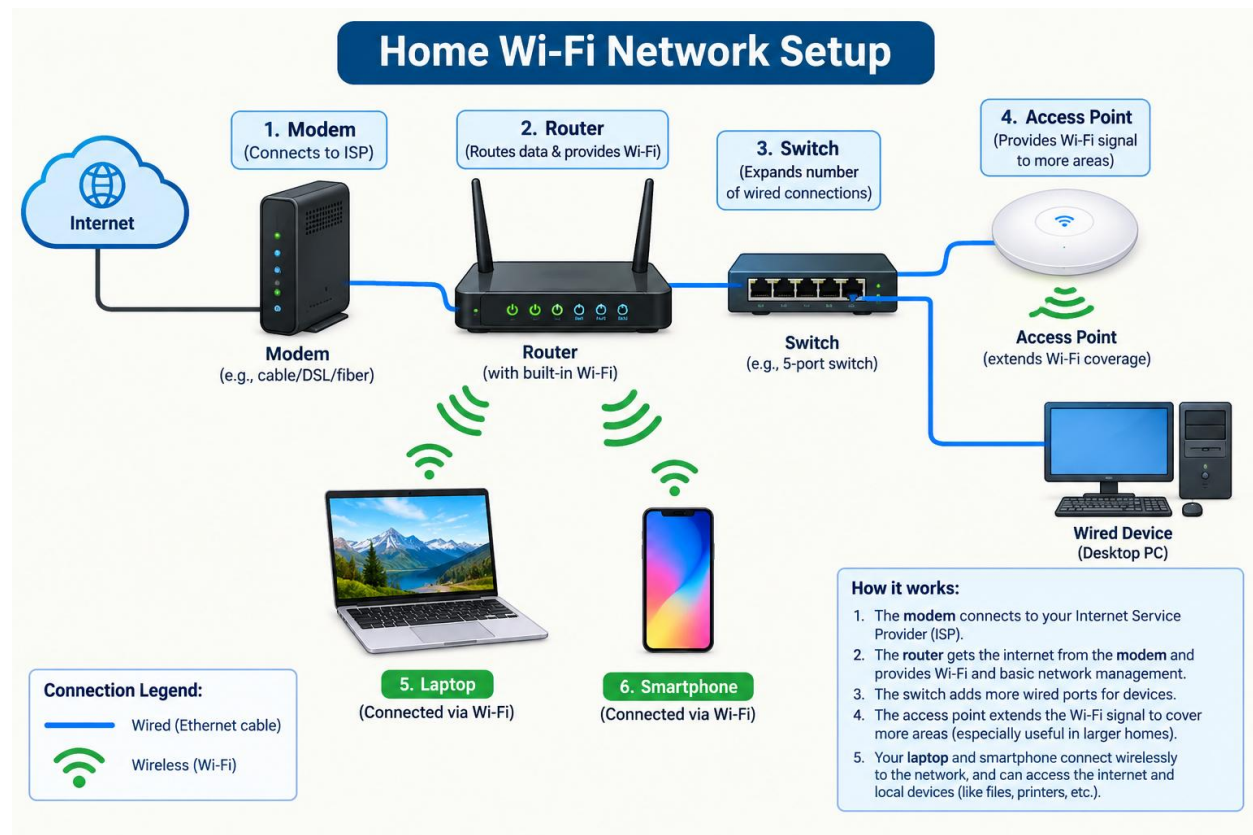


ASSIGNMENT -3

1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

Answer:-



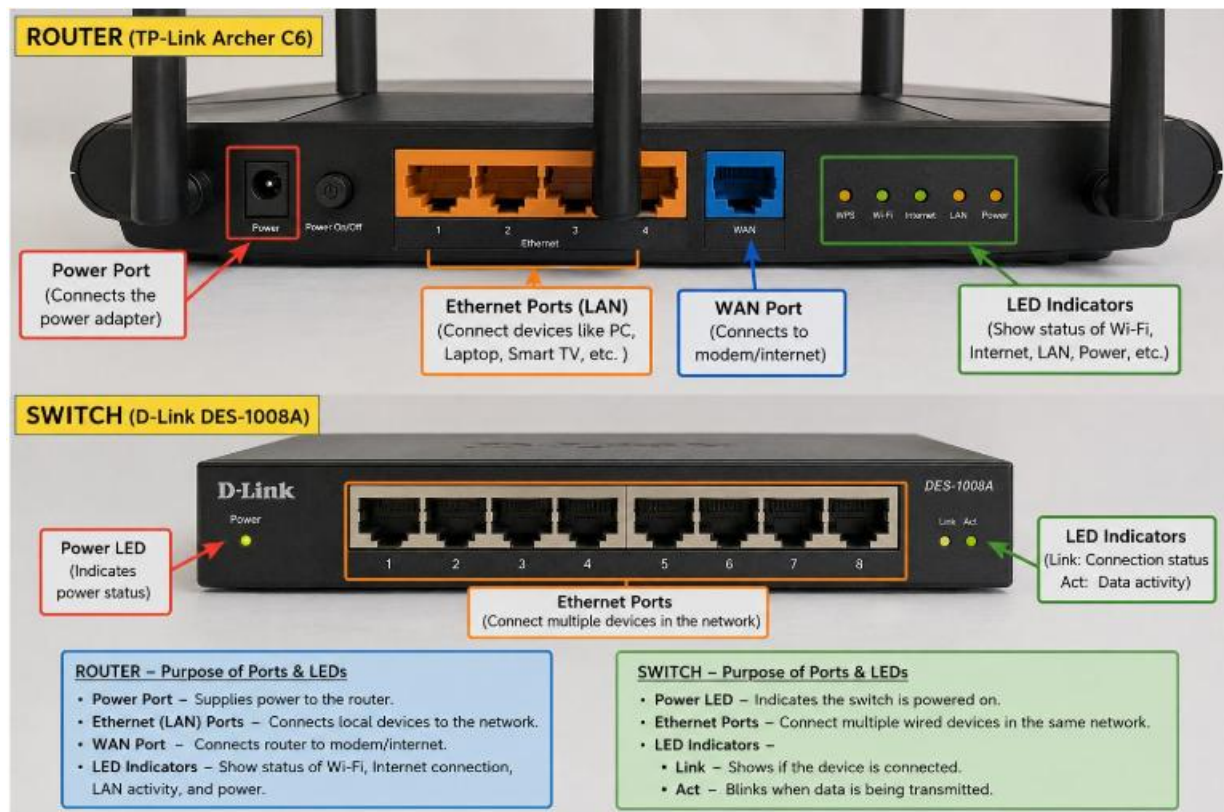
2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Answer:-

Networking Device	Main Function	Real-Life Example
Modem	Connects your home network to the Internet Service Provider (ISP).	Used at home to receive internet service from an ISP.
Router	Directs data between networks and shares the internet with multiple devices.	Used at home to connect phones, laptops, and TVs to the internet.
Switch	Connects multiple wired devices on the same local network.	Used in an office to connect several computers using Ethernet cables.
Access Point (AP)	Provides or extends Wi-Fi connectivity to wireless devices.	Used in a large office, college, or hotel to provide Wi-Fi in areas where the router's signal is weak.
Network Cable (Ethernet)	Carries data between network devices through a wired connection.	Used to connect a desktop PC, printer, or switch to a router.

3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

Answer:-



4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Answer:-

Scenario: Gaming Café Network

I am setting up a small gaming café for **IPL streaming and multiplayer gaming**. I would connect the **router first to the ISP line** because the router provides Internet access, assigns IP addresses, and manages traffic between the café network and the Internet.

Then, I would connect the router to an **8-port or 16-port network switch** using an Ethernet cable. The **8 gaming PCs** would each be connected to separate ports on the switch using Ethernet cables.

Network layout:

ISP → Router → Switch → 8 Gaming PCs

The **router** is needed to connect the café's local network to the Internet, while the **switch** allows all 8 PCs to communicate efficiently on the local network. A **wired Ethernet connection** is preferred for gaming because it provides a more stable and lower-latency connection than relying on Wi-Fi.